

Downloading Libraries

Titanic Dataset - Exploratory Data Analysis (EDA)

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Introduction

The Titanic dataset is one of the most popular datasets for beginners to practice EDA and machine learning.

In this project, we aim to uncover hidden patterns, relationships, and factors that influenced the survival of passengers onboard the Titanic.

```
In [48]: #Downloading the required libraries  
!pip install pandas matplotlib seaborn missingno
```

Requirement already satisfied: pandas in c:\users\dell\anaconda3\lib\site-packages (2.2.3)
Requirement already satisfied: matplotlib in c:\users\dell\anaconda3\lib\site-packages (3.8.0)
Requirement already satisfied: seaborn in c:\users\dell\anaconda3\lib\site-packages (0.12.2)
Requirement already satisfied: missingno in c:\users\dell\anaconda3\lib\site-packages (0.5.2)
Requirement already satisfied: numpy>=1.23.2 in c:\users\dell\anaconda3\lib\site-packages (from pandas) (1.26.4)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\dell\anaconda3\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\dell\anaconda3\lib\site-packages (from pandas) (2025.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\dell\anaconda3\lib\site-packages (from pandas) (2025.1)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\dell\anaconda3\lib\site-packages (from matplotlib) (1.2.0)
Requirement already satisfied: cycler>=0.10 in c:\users\dell\anaconda3\lib\site-packages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\dell\anaconda3\lib\site-packages (from matplotlib) (4.25.0)
Requirement already satisfied: kiwisolver>=1.0.1 in c:\users\dell\anaconda3\lib\site-packages (from matplotlib) (1.4.4)
Requirement already satisfied: packaging>=20.0 in c:\users\dell\anaconda3\lib\site-packages (from matplotlib) (23.2)
Requirement already satisfied: pillow>=6.2.0 in c:\users\dell\anaconda3\lib\site-packages (from matplotlib) (10.4.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\dell\anaconda3\lib\site-packages (from matplotlib) (3.0.9)
Requirement already satisfied: scipy in c:\users\dell\anaconda3\lib\site-packages (from missingno) (1.11.4)
Requirement already satisfied: six>=1.5 in c:\users\dell\anaconda3\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)

Importing Libraries

```
In [41]: #Importing the Libraries  
import pandas as pd  
import seaborn as sns  
import matplotlib.pyplot as plt  
import missingno as msno
```

Loading The Dataset

```
In [4]: #Loading the Dataset  
data = pd.read_csv("C:\\Users\\Dell\\Downloads\\Titanic-Dataset.csv")
```

Basic Exploration

```
In [5]: #Basic exploration  
print(data.head())
```

```
print(data.info())  
print(data.describe())
```

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 891 entries, 0 to 890

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	object
4	Sex	891 non-null	object
5	Age	714 non-null	float64
6	SibSp	891 non-null	int64
7	Parch	891 non-null	int64
8	Ticket	891 non-null	object
9	Fare	891 non-null	float64
10	Cabin	204 non-null	object
11	Embarked	889 non-null	object

dtypes: float64(2), int64(5), object(5)

memory usage: 83.7+ KB

None

	PassengerId	Survived	Pclass	Age	SibSp	\
count	891.000000	891.000000	891.000000	714.000000	891.000000	
mean	446.000000	0.383838	2.308642	29.699118	0.523008	
std	257.353842	0.486592	0.836071	14.526497	1.102743	
min	1.000000	0.000000	1.000000	0.420000	0.000000	
25%	223.500000	0.000000	2.000000	20.125000	0.000000	
50%	446.000000	0.000000	3.000000	28.000000	0.000000	
75%	668.500000	1.000000	3.000000	38.000000	1.000000	
max	891.000000	1.000000	3.000000	80.000000	8.000000	

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

Data Cleaning

```
In [6]: #Data Cleaning
#Checking for missing values
print(data.isnull().sum())
```

```
PassengerId      0
Survived          0
Pclass           0
Name             0
Sex              0
Age             177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin           687
Embarked         2
dtype: int64
```

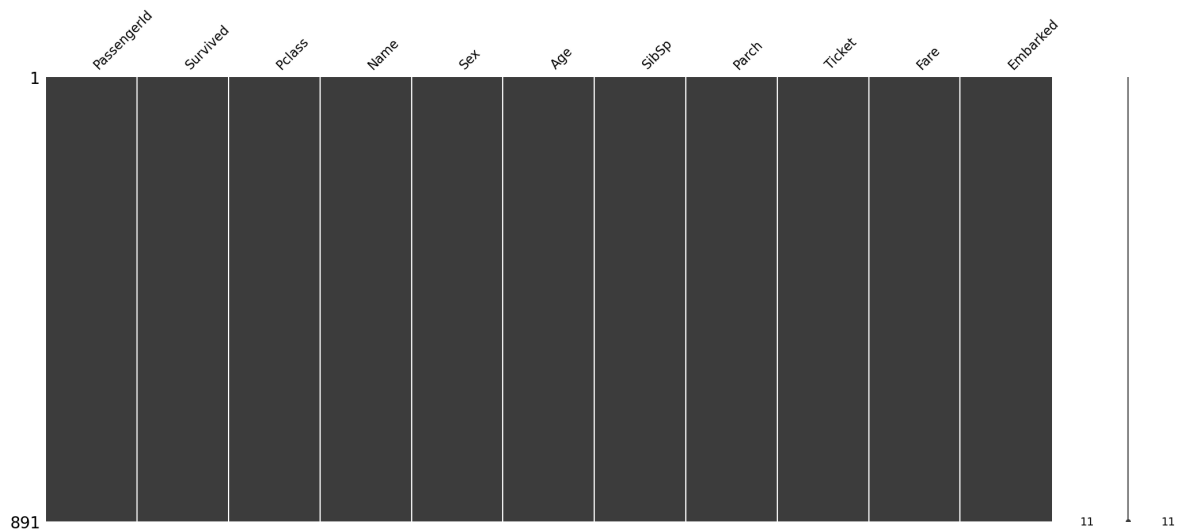
Handling The Missing values

```
In [11]: #Handling missing values
data['Age'].fillna(data['Age'].median())
data['Embarked'].fillna(data['Embarked'].mode()[0])
```

```
Out[11]: 0      S
1      C
2      S
3      S
4      S
..
886    S
887    S
888    S
889    C
890    Q
Name: Embarked, Length: 891, dtype: object
```

Visualizing Missing Values

```
In [42]: #Visualizing missing values
msno.matrix(data)
plt.show()
```



Observation:

- Most features have complete data.
- Some missing values might be present in 'Age' and 'Embarked' (if visible).
- Overall, the dataset is fairly clean.

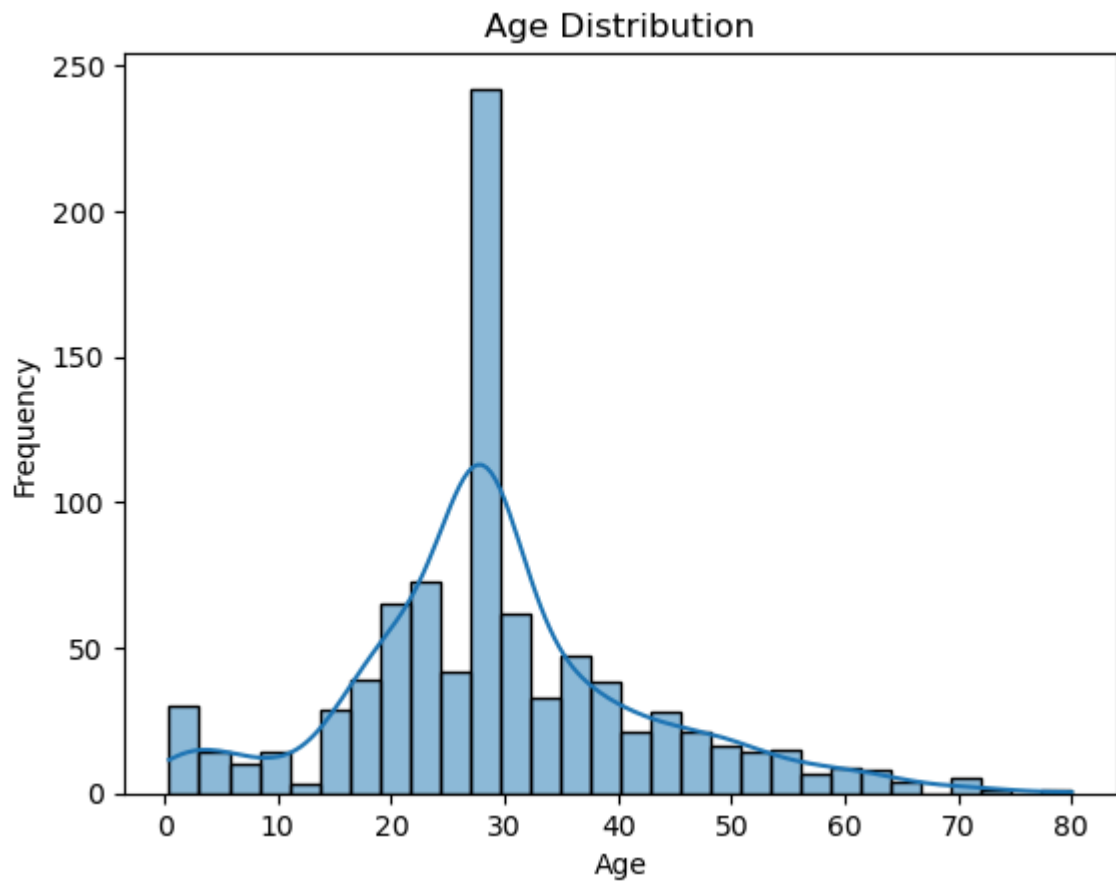
Univariate Analysis.

In this section, we will explore individual variables to understand their distributions and characteristics.

Exploring the distribution of passenger ages.

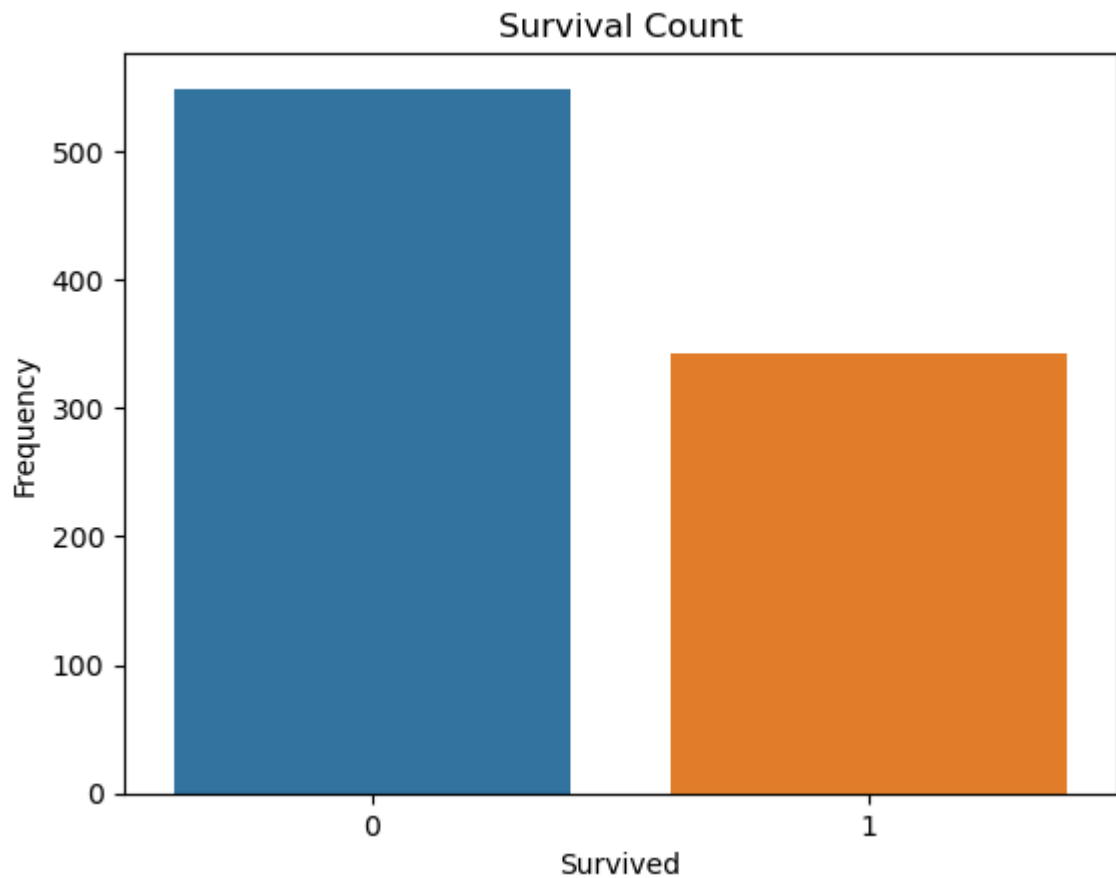
```
In [33]: #Univariate Analysis
#Histogram for Age Distribution
sns.histplot(data['Age'], kde= True)
plt.title('Age Distribution')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.show()
```

C:\Users\Dell\anaconda3\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.
with pd.option_context('mode.use_inf_as_na', True):

**Observation:**

- Most passengers were between 20 and 40 years old.
- Very few passengers were under 10 or over 60, showing a young adult-heavy passenger list.

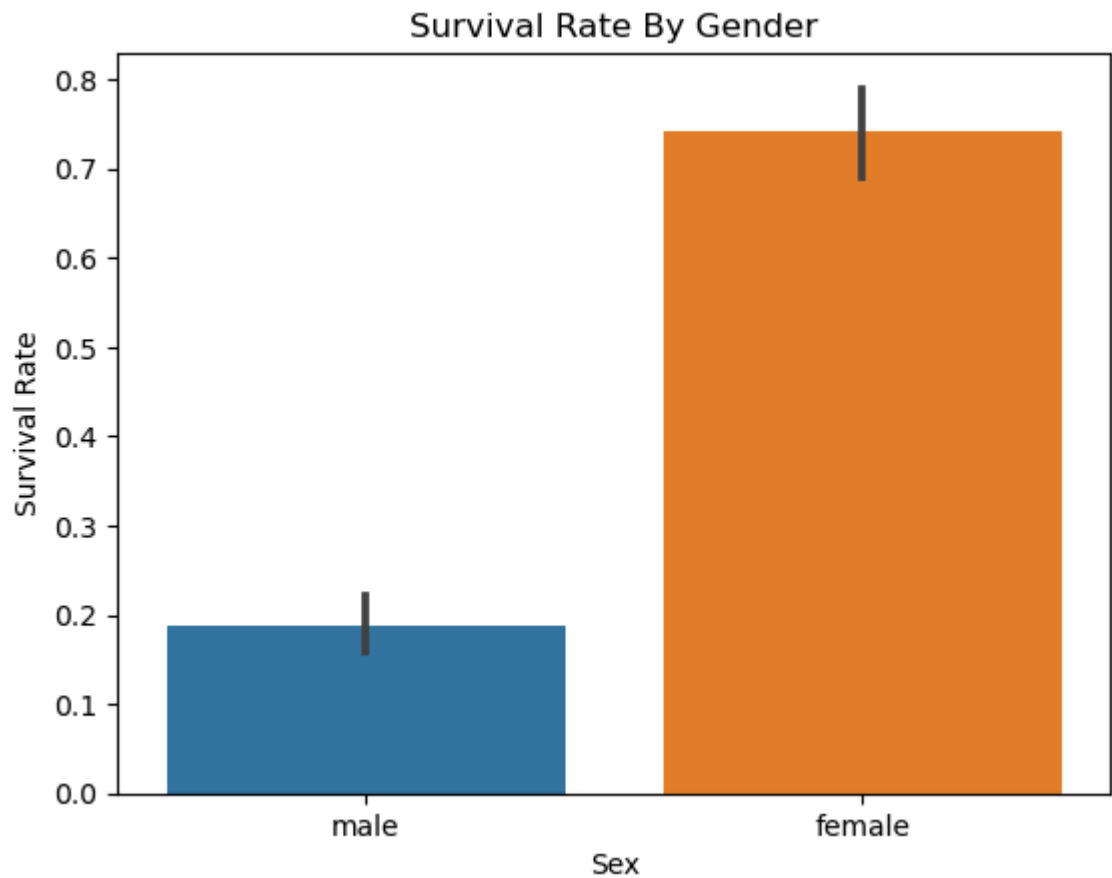
```
In [34]: #Countplot
#Survival Count
sns.countplot(x= 'Survived', data= data)
plt.title('Survival Count')
plt.xlabel('Survived')
plt.ylabel('Frequency')
plt.show()
```

**Observation:**

- Around 38% of passengers survived, while 62% did not.
- The dataset is slightly imbalanced toward non-survivors.

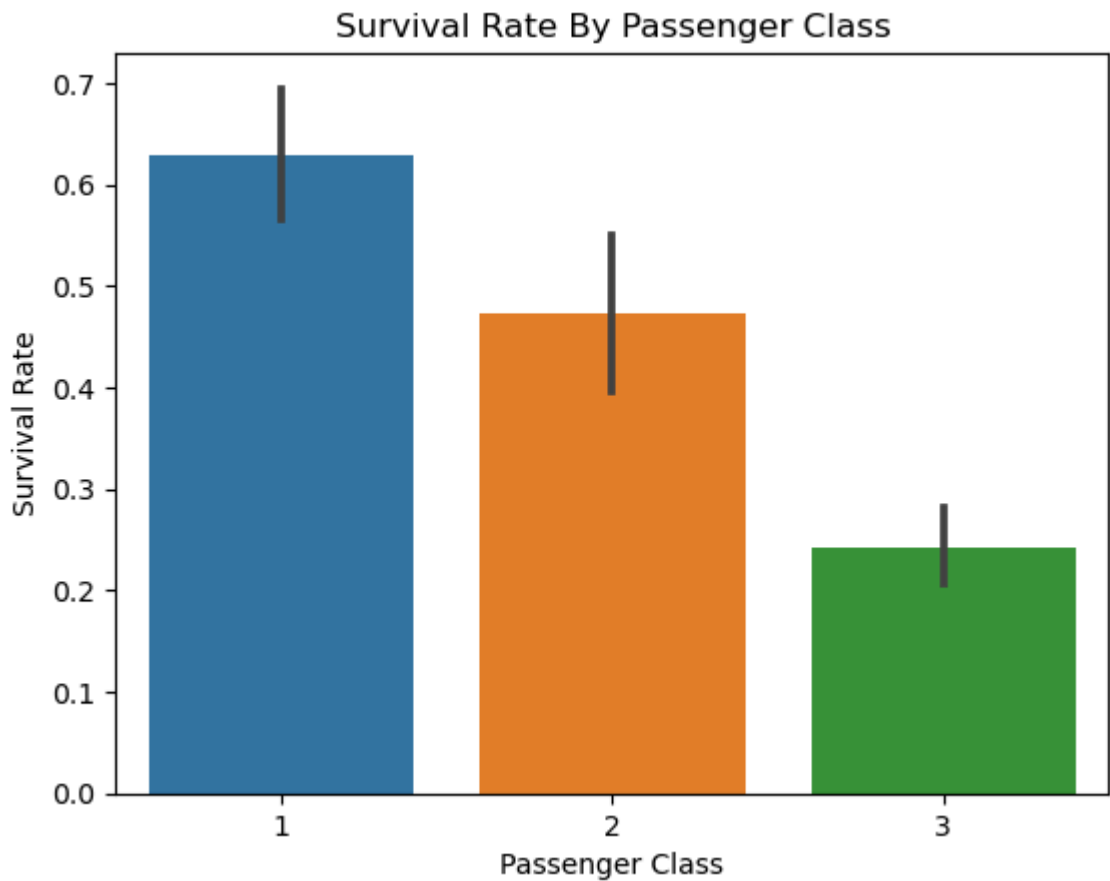
Bivariate Analysis

```
In [35]: #Bivariate Analysis
#Bar Plot
#Gender by Survival
sns.barplot(x= 'Sex', y='Survived', data=data)
plt.title('Survival Rate By Gender')
plt.xlabel('Sex')
plt.ylabel('Survival Rate')
plt.show()
```


**Observation:**

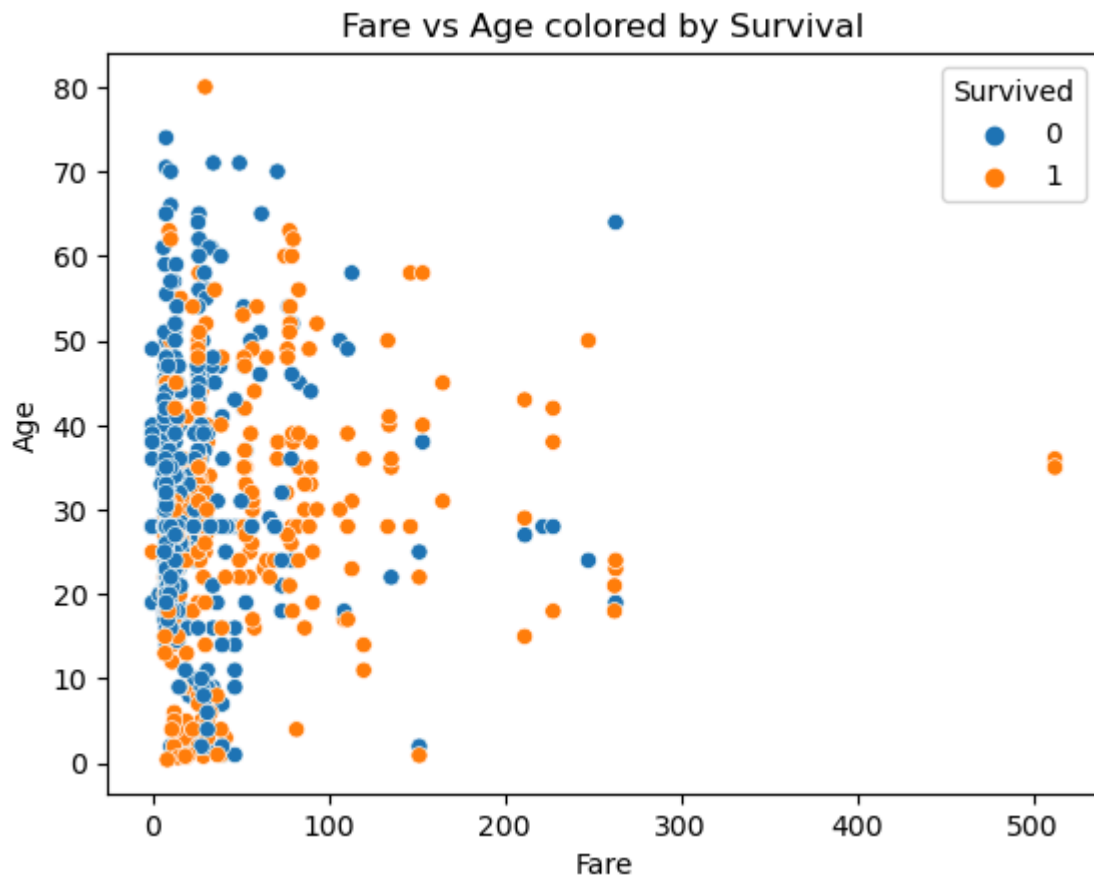
- A much higher percentage of females survived compared to males.
- Gender was a strong factor influencing survival chances on the Titanic.

```
In [36]: #Bar Plot2 for Survival by Passenger Class
sns.barplot(x='Pclass', y= 'Survived', data=data)
plt.title('Survival Rate By Passenger Class')
plt.xlabel('Passenger Class')
plt.ylabel('Survival Rate')
plt.show()
```

**Observation:**

- Passengers from 1st Class had significantly higher survival rates.
- Lower class passengers (especially 3rd class) faced lower survival chances.

```
In [32]: #Scatterplot for Fare Distribution
sns.scatterplot(x='Fare', y='Age', hue= 'Survived', data=data)
plt.title('Fare vs Age colored by Survival')
plt.show()
```

**Observation:**

- Passengers who paid higher fares generally had better survival chances.
- Younger passengers had varying survival rates across different fare amounts

Multivariate Analysis

```
In [37]: #Multivariate Analysis

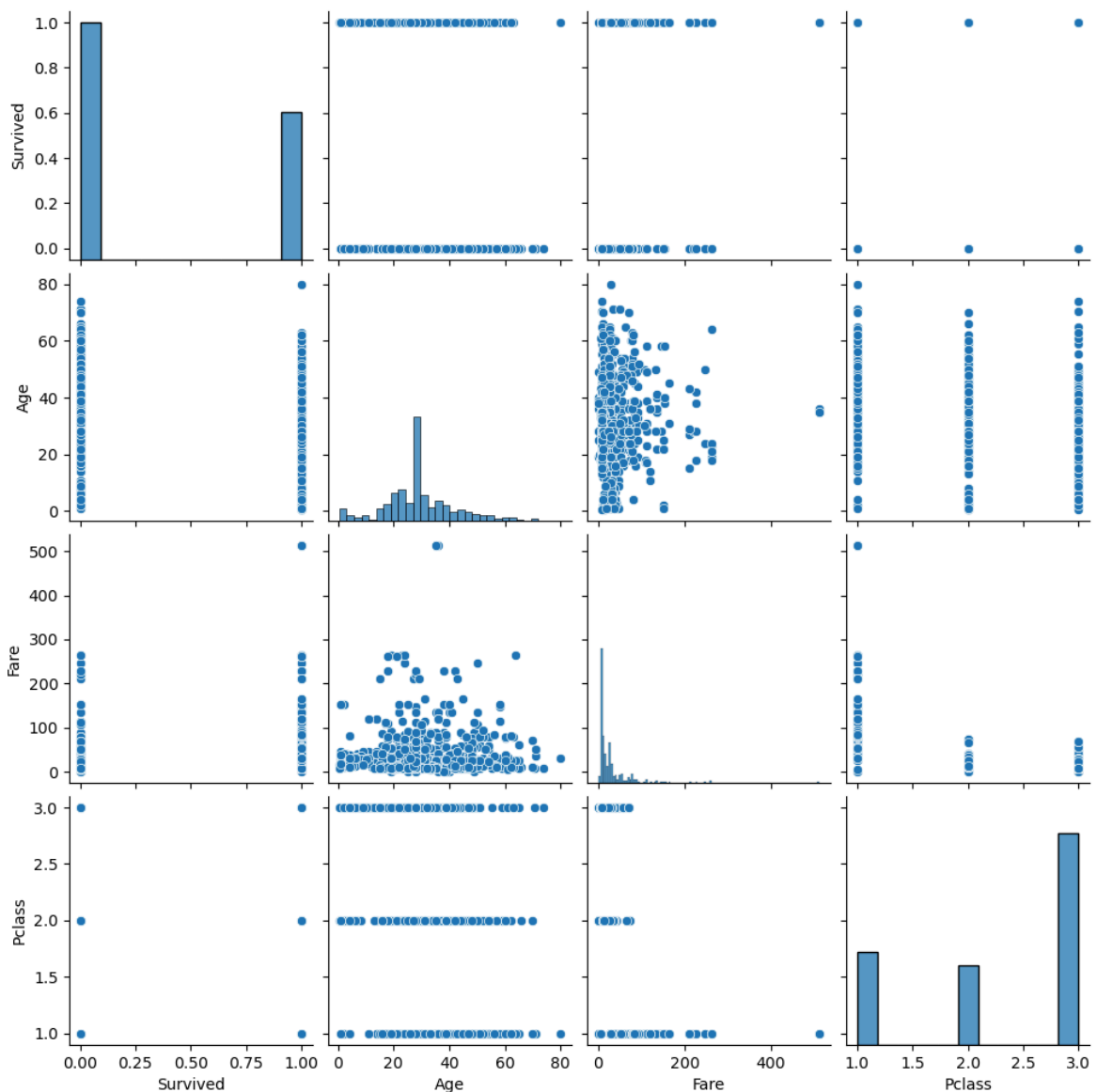
#Pairplot
sns.pairplot(data[['Survived', 'Age', 'Fare', 'Pclass']])
plt.show()

numeric_data= data.select_dtypes(include=['float64', 'int64'])
```

```

C:\Users\Dell\anaconda3\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Convert
inf values to NaN before operating instead.
  with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\anaconda3\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Convert
inf values to NaN before operating instead.
  with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\anaconda3\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Convert
inf values to NaN before operating instead.
  with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\anaconda3\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Convert
inf values to NaN before operating instead.
  with pd.option_context('mode.use_inf_as_na', True):

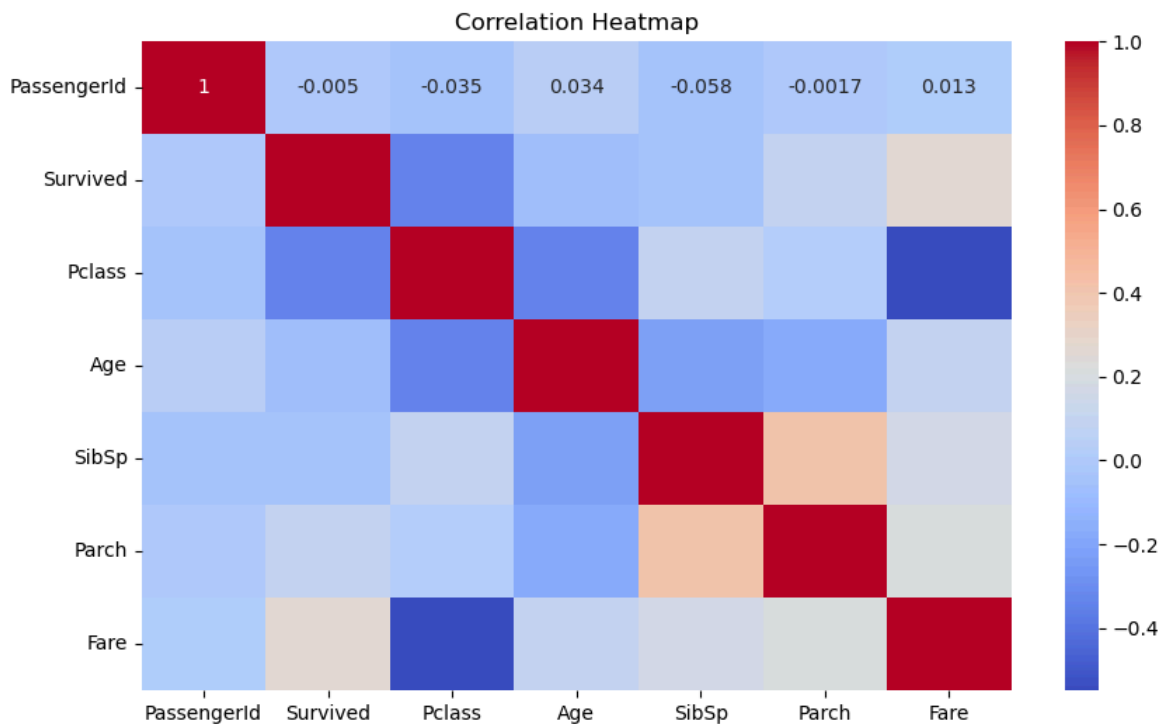
```



Observation:

- Clear visual separation is seen where higher class and higher fare passengers survived more.
- Younger first-class passengers especially had a better survival chance.

```
In [38]: #Heatmap
plt.figure(figsize=(10,6))
sns.heatmap(numeric_data.corr(),annot= True, cmap= 'coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```

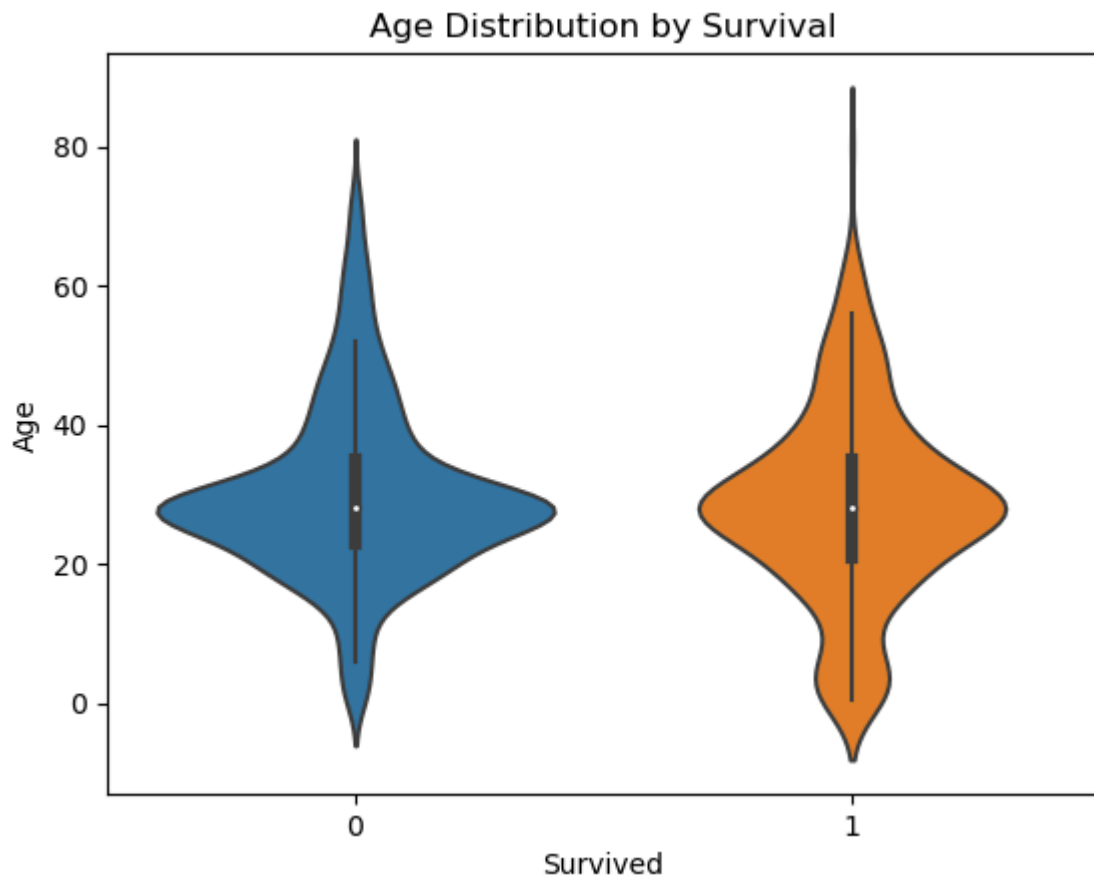


Obsevation:

- Fare and Pclass are strongly correlated with survival.
- Age and survival have a weaker correlation compared to class and fare.

Advanced Visualizations

```
In [44]: #Violinplot: Age distribution by Survival
sns.violinplot(x='Survived', y='Age', data=data)
plt.title('Age Distribution by Survival')
plt.xlabel('Survived')
plt.ylabel('Age')
plt.show()
```

**Observation:**

Younger passengers had a higher chance of survival.

Non-survivors were spread across all age groups, while survivors mostly were younger.

```
In [45]: # SwarmPlot: Fare by Gender
sns.swarmplot(x='Sex', y='Fare', data=data)
plt.title('Fare paid by Gender')
plt.xlabel('Gender')
plt.ylabel('Fare')
plt.show()
```

```

C:\Users\Dell\anaconda3\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.
  with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\anaconda3\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.
  with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\anaconda3\Lib\site-packages\seaborn\_oldcore.py:1075: FutureWarning: When grouping with a length-1 list-like, you will need to pass a length-1 tuple to get_group in a future version of pandas. Pass `(name,)` instead of `name` to silence this warning.
  data_subset = grouped_data.get_group(pd_key)
C:\Users\Dell\anaconda3\Lib\site-packages\seaborn\categorical.py:3544: UserWarning: 56.3% of the points cannot be placed; you may want to decrease the size of the markers or use stripplot.
  warnings.warn(msg, UserWarning)
C:\Users\Dell\anaconda3\Lib\site-packages\seaborn\categorical.py:3544: UserWarning: 28.7% of the points cannot be placed; you may want to decrease the size of the markers or use stripplot.
  warnings.warn(msg, UserWarning)
C:\Users\Dell\anaconda3\Lib\site-packages\seaborn\categorical.py:3544: UserWarning: 65.5% of the points cannot be placed; you may want to decrease the size of the markers or use stripplot.
  warnings.warn(msg, UserWarning)
C:\Users\Dell\anaconda3\Lib\site-packages\seaborn\categorical.py:3544: UserWarning: 42.7% of the points cannot be placed; you may want to decrease the size of the markers or use stripplot.
  warnings.warn(msg, UserWarning)

```



Observation:

Males generally paid slightly lower fares than females.

A few outliers paid extremely high fares regardless of gender.

Key Insights

- Most passengers who survived were in the age group of 20-40.
- Females had a much higher survival rate than males.
- Higher class passengers (1st class) had better survival chances.
- Families with more than 3 members were less likely to survive.
- Fare distribution is highly right-skewed, with a few paying exceptionally high fares.

Conclusion

The exploratory data analysis revealed important factors influencing survival rates on the Titanic.

References

- Dataset Source: [Kaggle Titanic Dataset](#)

In []: